INFORM Tanzania — Methodology & Data-Integrity Manual

INFORM Tanzania — Methodology & Data-Integrity Manual

Living document. Honest by design. This explains, end to end, how a raw measurement (a population count, a rainfall record, a flood event) becomes a 0–10 indicator, how indicators combine into the INFORM Risk, and how a Data-Entry edit travels the same path. Where a method is approximate, or a value looks too close to a neighbour, it is **flagged here openly** so we can improve it together — not hidden. Numbers below are real, taken from the current dataset (`src/data/tanzania-inform-risk.json`).

---

1. The risk journey (one picture)

```
 raw data ──standardize──▶ indicator 0–10 ──MEAN──▶ category 0–10
        └──────────────────────────────────────────────────┘
 category 0–10 ──scaled GEOMEAN (Excel Box 6)──▶ dimension 0–10
 Hazard&Exposure · Vulnerability · Lack-of-Coping ──∛(H×V×LCC)──▶ RISK 0–10 ──▶
class
```

Three rules, taken **exactly from the Tanzania INFORM Excel** (`TZ_INFORM_model.xlsx` , locked by `excelParity.test.js`):

Step	Method	Where
indicator → category	**arithmetic MEAN**	`AVERAGEIFS` in Excel
category → dimension	**INFORM scaled geometric mean**	Excel Box 6 (below)
dimension → risk	**$\sqrt[3]{(H \times V \times LCC)}$** (geometric mean)	INFORM Risk index

The same three rules run in the offline build (`scripts/apply-climate-hazards.mjs`) **and** live in the app when you edit (`riskModel.applyEdits`) — so a computed value and an edited value behave identically.

---

2. Standardization: turning a raw number into 0–10

Every indicator must sit on the same 0–10 scale ("higher = more risk") before it can be combined. We use **min-max** scaling across Tanzania's districts:

> **idx = (value − min) / (max − min) × 10**

For quantities that are highly skewed (a few huge cities, many small rural areas) we take the **log first** (log min-max), so the scale is not dominated by one outlier:

> **idx = ($\log_{10}$value − $\log_{10}$min) / ($\log_{10}$max − $\log_{10}$min) × 10**

Worked example — population → Exposure (your "3900" question)

Exposure is **population density** (people per km²), log-scaled because Dar (4,600/km²) dwarfs pastoral Longido (22/km²).

- Ilala (Dar): density 4,643/km² → $\log_{10}$ = 3.667 → **exposure 8.66**
- Kondoa: density 54/km² → $\log_{10}$ = 1.73 → **exposure 3.19**
- Longido: density 22/km² → $\log_{10}$ = 1.34 → **exposure 2.10**

So a district of (say) 3,900 people in a 1,300 km² ward = density 3.0/km² → near the bottom of the scale; the *same* 3,900 people in 5 km² = 780/km² → high. **Exposure is about concentration, not the raw headcount.** (Population comes from the NBS 2022 census, per council, summed to the district — see §7.)

Why min-max and not "rank"

An earlier draft ranked districts (1st, 2nd, …). That spreads everything evenly but **invents differences where the data is flat** — it once put random districts top of the drought list. Min-max keeps the *real* spacing: if two districts are genuinely similar, their scores stay close (and yes, sometimes they look "too close" — that is honest, see §8).

---

3. Hazard indicators — what we compute, honestly

3.1 Drought (computed — CHIRPS v3 + ERA5)

Drought is **slow** and hits communities through **failed growing seasons** (→ food insecurity, IPC/MUCHALI). SPI/SPEI *frequency* is ~16 % everywhere by construction, so it can NOT rank where drought is worse. Instead we blend four real signals, each min-maxed:

> **drought = minmax(0.30·aridity + 0.20·variability + 0.15·SPEI-depth + 0.35·season-failure) × 10**

- **aridity** = low mean annual rainfall (1991–2024 CHIRPS v3)
- **variability** = interannual rainfall CV
- **SPEI-depth** = mean magnitude of SPEI-12 during droughts (uses ERA5 temperature → Thornthwaite PET)
- **season-failure** = how often the main 4-month rainy season falls below 80 % of normal

Result (top): Longido 9.7, Simanjiro 9.1, Singida 8.7, Hanang 8.7, Kondoa 8.6 — the real pastoral north + semi-arid centre. Bottom: Rungwe 0.5, Kyela 0.5, Mafia 1.5 — the wet highlands/islands. *(Script: `scripts/compute-drought-spi-spei.py`.)*

The computed value is **floored at the documented baseline** (`drought = max(computed, documented)`) — like flood, a known drought is never lowered, only raised where the climate evidence is stronger.

3.2 Heavy rainfall & flood (computed — CHIRPS daily + recorded events)

Two ingredients: 1. **Heavy-rain index** — count of days where the district-mean rainfall exceeds **50 mm** (heavy) and **100 mm** (very heavy), per year, 2015–2024 daily CHIRPS, min-maxed (very-heavy weighted ×2). Top: the Pemba/Zanzibar coast (Mkoani 14 heavy days/yr). 2. **Recorded floods** — a documented, *editable* event list (`data-source/flood_events.csv`) with sources (DesInventar / EM-DAT / PMO-DMD / news). `event index = min(10, 4 + 2 × #events)`.

The **physical flood hazard** = `max(heavy-rain index, event index)` — so a coastal cloudburst zone *or* a documented river-flood district both score high.

3.3 Flood = Hazard, amplified by Exposure (never lowered)

INFORM hazard is **Hazard × Exposure** — but exposure must only ever **raise** a flood score, never hide a flood-prone place. So:

> **hazardFlood = max(documented flood , heavy-rain index , recorded-event index) flood = max(hazardFlood , √(hazardFlood × exposure))**

Exposure lifts the score only where population *exceeds* the hazard (dense flood cities); a sparsely-populated but flood-prone district keeps its full documented hazard.

- **Ilala (Dar):** heavy-rain 4.0, 2 recorded floods (2011, 2018) → event index 8 → hazardFlood = **8**. Exposure 8.66 → √(8 × 8.66) = 8.3 → flood = max(8, 8.3) = **8.3**.
- **Pangani (coastal, flood-prone but sparse):** documented hazard 9.8, exposure 2.9 → √(9.8 × 2.9) = 5.3 → flood = max(9.8, 5.3) = **9.8 — preserved.**

> ⚠️ **An earlier draft multiplied flood *down* by exposure (Pangani 9.8 → 3.3, Korogwe 8 → 2.1 — hiding known flood areas). That bug was caught by a before/after audit and fixed: exposure now amplifies only; a documented flood hazard is never lowered.**

3.4 Fires, earthquake, landslide, coastal — honest status

- **Wildfire, earthquake, lightning, volcano, storms, environmental degradation** are still the **INFORM SADC 2024 baseline** values (not yet recomputed here). We did **not** fabricate these. Wildfire, e.g., should later come from MODIS/VIIRS burned-area — flagged in §8.
- **Landslide & coastal** keep their **documented** values (Hanang, Rungwe, Usambara; coastal strip) — real but qualitative, editable, not yet a physical model.

---

4. Vulnerability & Coping — current sources

- **Development & poverty** — NBS Household Budget Survey 2017/18 (poverty headcount).
- **Children health & nutrition** — TDHS 2022 (under-5 stunting).
- **Livelihoods / food security** — wired to **MUCHALI / IPC** (MoA Food Security & Nutrition analysis). *We do not invent IPC phases;* the slot is ready for MoA to enter each round (§6).
- **WASH** — derived from **real 2022-census water points & boreholes** (more resource → lower “lack of coping”), rank-scaled 2–8.
- **Access to health / education** — 2022 PHC facility counts.
- **DRR implementation** — EPRP / regional EOCC / drought Anticipatory-Action records (the districts you provided) lower the institutional “lack of coping”.
- Indicators not yet localized keep the INFORM SADC 2024 baseline.

---

5. Aggregation — the exact formulas

5.1 Indicator → Category: MEAN

`category = average of its indicators` (ignoring blanks). E.g. Kondoa Natural-hazards = mean(drought 8.6, flood 3.8, earthquake 10, landslide 4.7, wildfire 4.75, …) = **4.0**.

5.2 Category → Dimension: INFORM scaled geometric mean (Excel Box 6)

> **dimension = (10 − GEOMEAN($(10-c_1)/10 \cdot 9 + 1$, $(10-c_2)/10 \cdot 9 + 1$, …)) / 9 × 10**

A low category **drags the dimension down** (geometric), unlike a plain average.

Worked example — Kondoa Hazard: Natural 4.0, Human 0.7. scaled: $(10-4.0)/10 \cdot 9+1 = 6.4$; $(10-0.7)/10 \cdot 9+1 = 9.37$. GEOMEAN = $\sqrt{6.4 \times 9.37} = 7.74$. dimension = $(10 - 7.74)/9 \times 10$ = **2.5**. ✔ matches stored.

5.3 Dimension → Risk: cube root

> **Risk = ∛(Hazard × Vulnerability × LackOfCoping)**

Kondoa: $\sqrt[3]{2.5 \times 5.5 \times 4.6} = \sqrt[3]{63.3}$ = **4.0** (Medium). ✔

Then `Risk` is classed with the **Tanzania thresholds** (Very-Low < 2.5 < Low < 3.4 < Medium < 4.3 < High < 5.9 < Very-High).

---

6. Keying data in (Data Entry) — same formulas, with provenance

1. Pick **role** (PMO/Admin apply directly; Sector officer submits for approval), region, district.
2. Edit a **dimension score** directly, or **Fill indicators** to set each 0–10 leaf — for **Hazard, Exposure, Vulnerability, Coping**.
3. **Exposure** has its own box; changing it instantly recomputes flood = √(hazard × exposure) and the Hazard score — the live preview uses the *same* mean → scaled-geomean → ∛ chain.
4. Choose the **Data source / authority** (NBS, TMA, MoW, MoH, MoA, PMO-DMD, DRR Coordinator, MUCHALI/IPC). Each edited indicator is **stamped** with that authority + date.
5. **Save / Submit → Apply.** The edit flows to the map, charts, tables. Hover the indicator and it shows **"✎ Updated by MoW · 2026-06"**; un-edited indicators show the registry source (e.g. **"📎 Source: TMA (+MoA, CHC) · CHIRPS v3 …"**).

Storage today is the browser (`localStorage`); the Supabase schema is ready for multi-user.

---

7. Data sources (per indicator)

Indicator	Authority	Dataset	How standardized
Drought	TMA / MoA	CHIRPS v3 1991-2024 + ERA5 t2m	aridity+variability+SPEI-depth+season-fail, min-max
Flood	PMO-DMD / TMA / MoW	CHIRPS v3 daily + recorded floods	max(>50/100 mm count, events) × exposure
Exposure	NBS	2022 PHC population	density, **log** min-max
Poverty	NBS	HBS 2017/18	INFORM baseline
Stunting	MoH	TDHS 2022	INFORM baseline
Food security	MUCHALI / IPC (MoA)	IPC rounds	*to be entered*
WASH	MoW	2022 PHC water/boreholes	resource availability 2–8
Health/Education access	MoH / NBS	2022 PHC facilities	resource availability
DRR implementation	PMO-DMD / DRR Coordinator	EPRP/EOCC/AA records	investment overlay
Wildfire, earthquake, etc.	INFORM SADC 2024	INFORM	baseline (not yet recomputed)

---

8. Honest limitations — the “to-improve” list (flag & discuss)

1. **Drought exposure is agricultural, not urban.** We multiply *flood* by population exposure but **not drought** — multiplying drought by urban density would wrongly demote pastoral communities (Longido). Better: cropland/livestock exposure. *Open.*
2. **Heavy-rain uses CHIRPS v2.0 daily** (on-disk, 2015–2024) right now; the national **v3** daily is downloading and will be swapped in. Extreme-day counts are near-identical, but it is not yet the v3 we use for drought. *In progress.*
3. **Some values sit close together** (e.g. several semi-arid districts ~8.4–8.7 drought). That is real similarity, not precision — min-max preserves true spacing rather than forcing a spread. Don’t over-read 0.1 differences.

4. **Dar es Salaam drought = 10 (documented artifact).** Ilala/Temeke carry drought 10 from the documented INFORM baseline — clearly too high for a coastal city. The precautionary floor keeps documented values rather than silently lowering them, so it persists (the *computed* value is ~6.6). **Flagged for correction via Data Entry** (MoA/TMA), which now overrides it.
5. **Coverage: 138 / 170 districts** got computed climate (those with map polygons). The other 32 keep their prior values. Boundary set is 150/170 vs the 184 councils (2022) — reconciling to 184 needs the NBS council boundaries + council-level INFORM data.
6. **Wildfire, earthquake, lightning, storms** are still INFORM baseline — not yet physically recomputed (MODIS/VIIRS fire, USGS seismicity are the next sources).
7. **Standardization is relative to Tanzania** (min over our 150 districts), not INFORM's fixed global reference values. Fine for a national tool; note when comparing to global INFORM.
8. **Precautionary by design.** Flood and drought are *floored* at the documented baseline — computed evidence can **raise** a hazard but never **lower** it. This errs toward flagging risk (safer for planning) at the cost of keeping a few possibly-overstated documented values (see #4). A before/after audit confirms **0 districts** had flood or risk-class lowered.

Everything in this list is editable in Data Entry and improvable in code — that is the point of writing it down.

---

9. Risk over time (snapshots)

Risk is reported **"as of"** a date (`metadata.asOf`, currently the baseline). The plan to turn this into a time series — dated snapshots, **emerging** (risk ↑) vs **improving** (risk ↓) detection, and a Δrisk map — is in **RISK_OVER_TIME_PLAN.md**. Foundation already in place: every Data-Entry edit is timestamped with its authority, the CHIRPS inputs are inherently temporal (1991–2024), and the trend-chart components exist — so future seasonal refreshes can be compared to spot communities whose risk is rising before it becomes a disaster.